

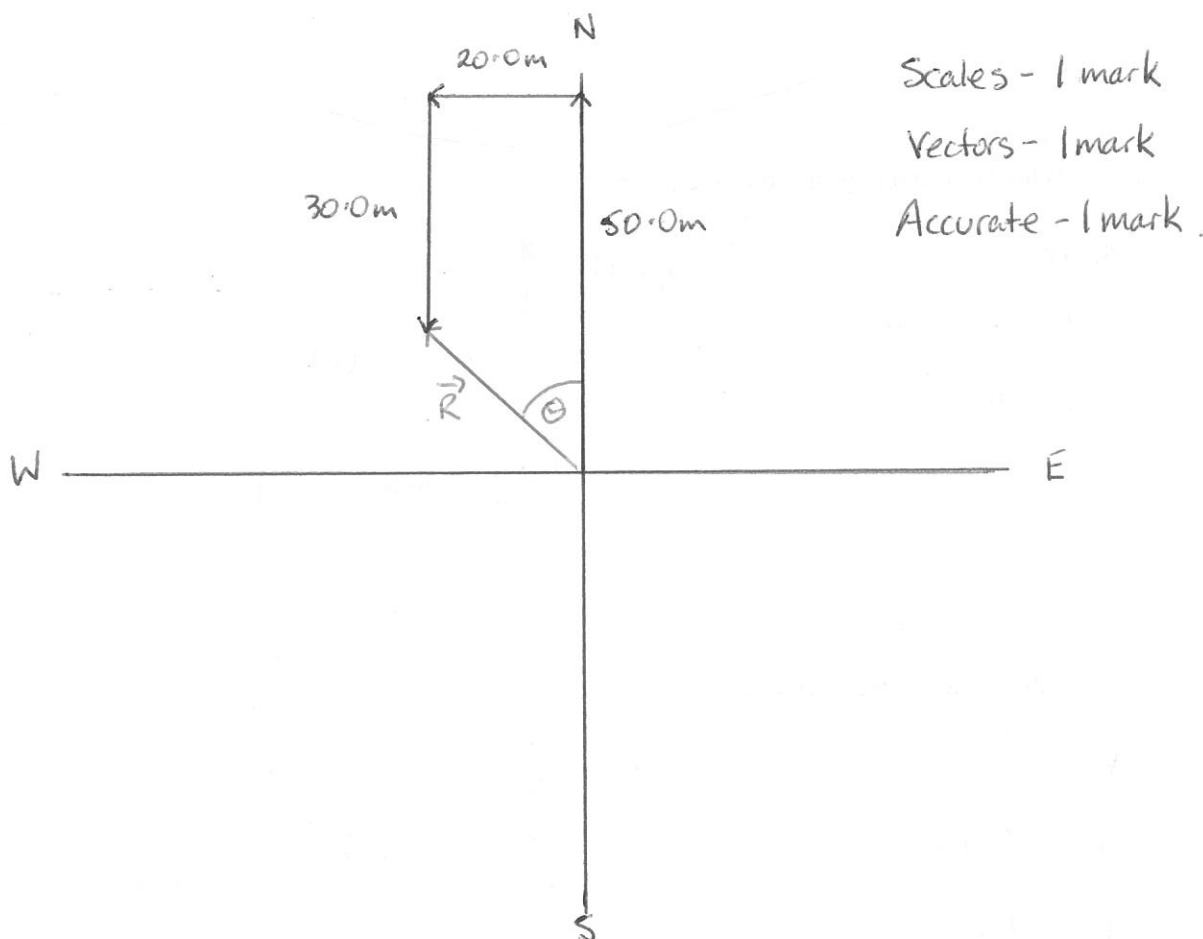
YEAR 10 SCIENCE (MODIFIED)
PHYSICAL SCIENCES
ASSIGNMENT - MOTION + NEWTON'S LAWS

NAME: SOLUTIONSMARK: 43

Apply relationships between force, mass and acceleration to predict changes in the motion of objects.

Mark	ND	NW	C	HC	O
Mark Range	0-12	13-18	19-26	27-31	32-43

1. On the school oval, a person runs 50.0 m North, then 20.0 m West and then 30.0 m South. The total time taken for the motion is 40.0 s.
- (a) Draw a vector diagram to show the motion. Draw in the resultant vector (use a different colour). (Use a scale of 1.0 cm = 10.0 m).



(3)

- (b) Calculate the distance covered.

$$d = 50.0 + 20.0 + 30.0 \quad (1)$$

$$= \underline{100.0 \text{ m}} \quad (1)$$

(2)

- (c) Using a ruler and protractor, determine the displacement of the person.

$$\vec{r} = 28 \text{ m} \quad (1)$$

$$\theta = 50^\circ \quad (1)$$

$$\therefore \underline{\vec{r} = 28 \text{ m N } 50^\circ \text{ W}} \quad (1)$$

(3)

- (d) What is the average speed of the person?

$$\text{Speed} = ?$$

$$d = 100.0 \text{ m}$$

$$t = 40.0 \text{ s}$$

$$\text{Speed} = \frac{d}{t}$$

$$= \frac{100.0}{40.0} \quad (1)$$

$$= \underline{2.50 \text{ ms}^{-1}} \quad (1)$$

(2)

- (e) What is the average velocity?

$$V = ?$$

$$s = 28 \text{ m}$$

$$t = 40.0 \text{ s}$$

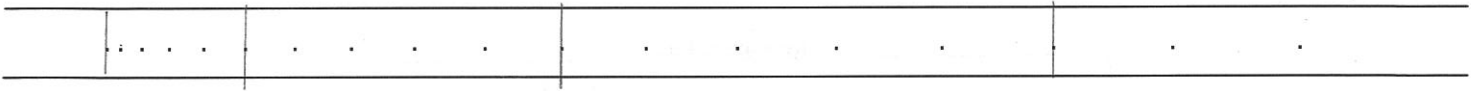
$$V = \frac{s}{t}$$

$$= \frac{28}{40.0} \quad (1)$$

$$= \underline{0.7 \text{ ms}^{-1} \text{ N } 50^\circ \text{ W}} \quad (1)$$

(3)

2. A group of students investigated the movement of a toy car down an inclined plane in the laboratory, and produced the following ticker-timer tape. The timer produces 50 dots per second.



- (a) Analyse the tape by calculating the velocity for every 5 dot-spacings. Record your values in the table below.

Distance (m)	Time (s)	Velocity (ms^{-1})
0.0	0.0	0.0
0.0185	0.1	0.185
0.042	0.1	0.42
0.065	0.1	0.65

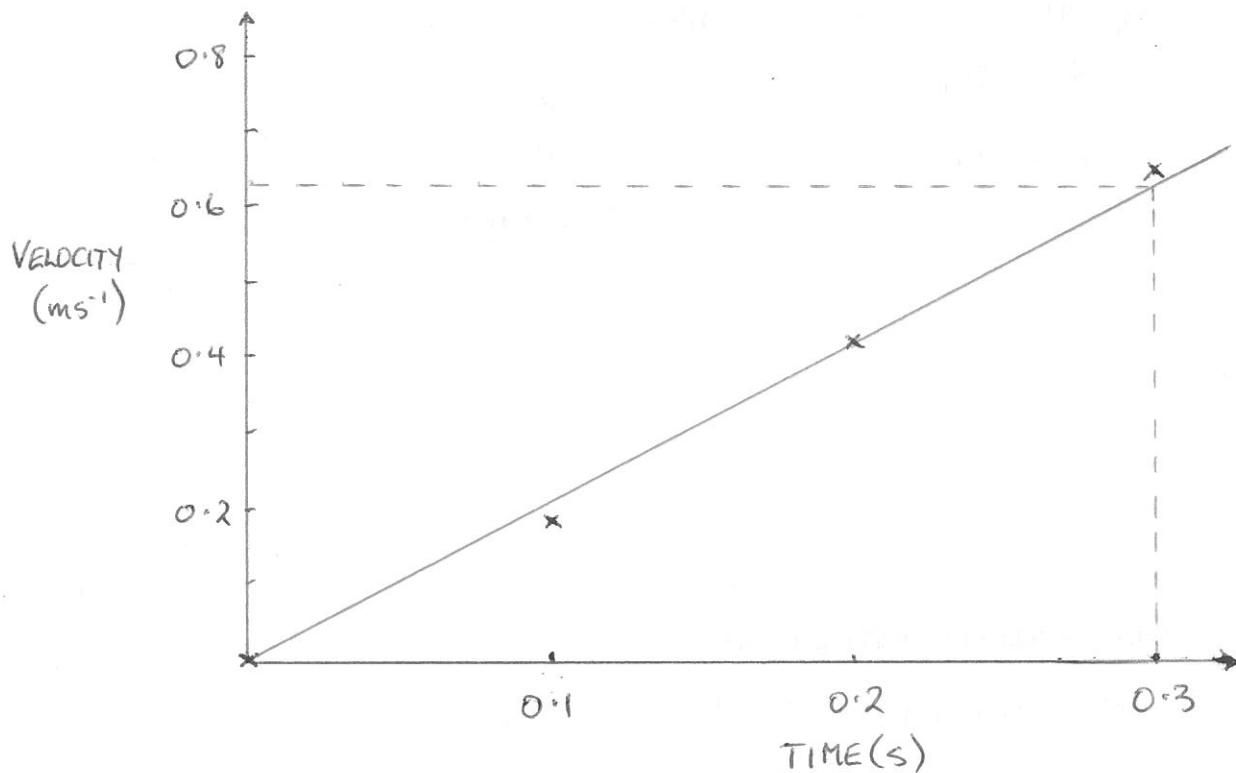
(1)

(1)

(1)

(3)

- (b) Draw a velocity - time graph for these measurements.



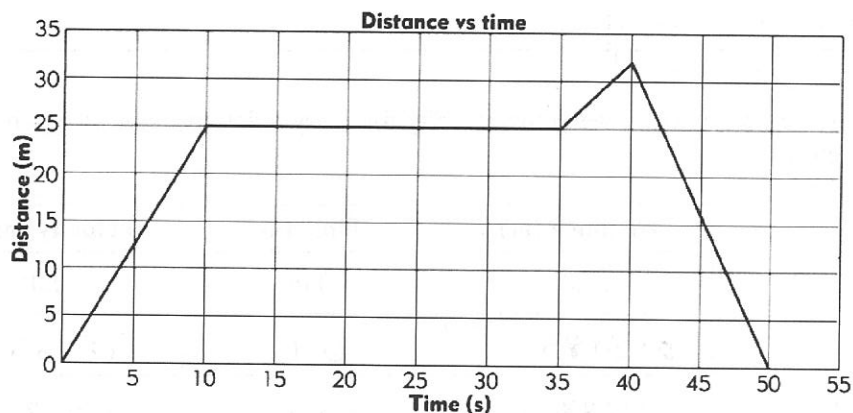
(3)

Labels + units - 1 mark

Scales - 1 mark

Plotting - 1 mark.

3. John decided to take a walk in the playground. He plotted the graph below to show how his distance changed with time during the walk.



- (a) At what time did John stop walking and talk with Brianna? 10 s (1)
- (b) How many seconds did their conversation take? 25 s (1)
- (c) Calculate John's speed during the first 10 s. Show your working.

$$\text{speed} = ?$$

$$d = 25 \text{ m}$$

$$t = 10 \text{ s}$$

$$\text{speed} = \frac{d}{t} \quad (1)$$

$$= \frac{25}{10} \quad (1)$$

$$= \underline{2.5 \text{ ms}^{-1}} \quad (1)$$

(3)

- (d) What was John's total walking distance?

$$d = 25 + 7 + 32 \quad (1)$$

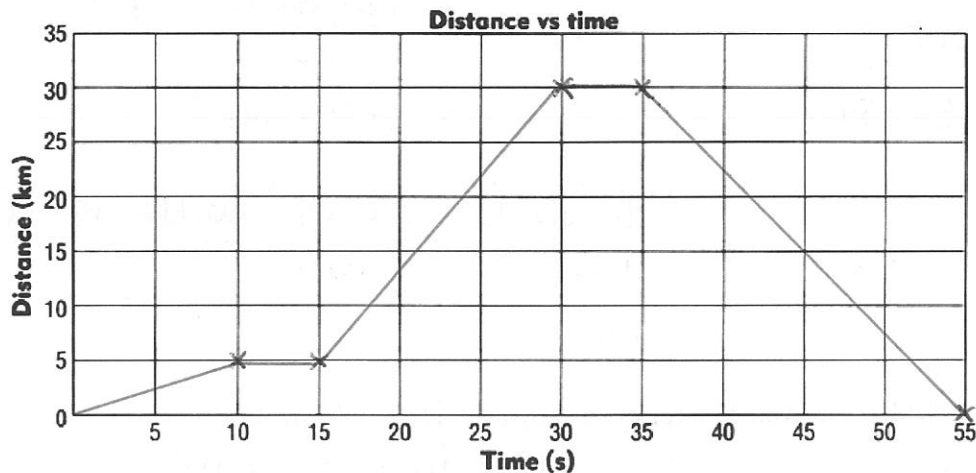
$$= \underline{64 \text{ m}} \quad (1)$$

(2)

4. Sketch a graph for the following journey on the axes below.

Glenn had to deliver a parcel to Inga. he left home and drove 5 km to fill up with petrol. It took him 10 minutes to reach the service station. Glenn stopped for 5 minutes to fill the car with petrol. He continued on his journey away from his starting point to Inga's house, and at a distance of 30 km he arrived. This part of the journey took another 15 minutes. Glenn talked with Inga for 5 minutes before getting back into his car and returning home. The trip back home took Glenn 20 minutes, during which time he did not stop.

(3)



Shape - 1 mark
Accuracy - 2 marks

5. Complete the table below by calculating the missing values.

Force (N)	Mass (kg)	Acceleration (ms^{-2})
20.0	10.0	2.0
56.0	7.0	8.0
108	12.0	9.0

1 mark each.

(3)

6. A car engine supplies $4.50 \times 10^3 \text{ N}$ of force through the front wheels of a car and accelerates it uniformly from rest. The mass of the car is $1.40 \times 10^3 \text{ kg}$. Calculate the acceleration of the car.

$$F = 4500 \text{ N}$$

$$m = 1400 \text{ kg}$$

$$a = ?$$

$$a = \frac{F}{m} \quad (1) \quad \text{Forwards +ve.}$$

$$= \frac{4500}{1400} \quad (1)$$

$$= \underline{3.21 \text{ ms}^{-2} \text{ forwards}} \quad (1)$$

(3)

7. A truck of mass 15,000 kg is moving at 25.0 ms^{-1} along Roe Highway when the driver notices a small car that pulled out from a petrol station 60.0 m ahead. He hit the brakes sharply and just manages to avoid hitting the back of the car. The truck ended up moving at 10.0 ms^{-1} and took 5.0 s to achieve this.

(a) What is the deceleration of the truck?

Take forwards as +ve.

$$V = 10.0 \text{ ms}^{-1}$$

$$u = 25.0 \text{ ms}^{-1}$$

$$a = ?$$

$$t = 5.0 \text{ s}$$

$$a = \frac{V - u}{t} \quad (1)$$

$$= \frac{10.0 - 25.0}{5.0} \quad (1)$$

$$= -3.00 \text{ ms}^{-2}$$

$$\therefore \text{deceleration} = 3.0 \text{ ms}^{-2} \text{ backwards} \quad (1)$$

(3)

(b) Calculate the average force exerted by the brakes of the truck.

$$F = ?$$

Take forwards as +ve.

$$m = 15,000 \text{ kg}$$

$$a = -3.00 \text{ ms}^{-2}$$

$$F = ma \quad (1)$$

$$= (15,000)(-3.00) \quad (1)$$

$$= -45,000 \text{ N}$$

$$\therefore \text{Force} = 45,000 \text{ N backwards.} \quad (1)$$

(3)

8. Explain why you "lurch forward" when standing on a bus or train and the vehicle brakes suddenly.

- You have inertia since you are moving. (1)
- As the vehicle slows, your inertia carries you forwards. (1)

(2)